STA2023 – Statistical Methods I
Instructor: Dr. Emil Agbemade

Chapter 1. Data Collection

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Learning Objectives:
By the end of this chapter, you should know how to:

- Define statistics and explain the statistical process.
- Distinguish between different variable types.
- Distinguish between observational studies and designed experiments.
- Identify how a simple random sample is taken.
- Identify bias in sampling.

1.1 Introduction to the Practice of Statistics

- Whenever you take an introductory course, the first thing you must do is define the subject you are studying!

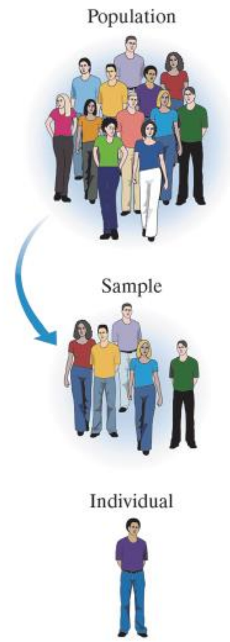
Definition: Statistics

Definition: Data

Statistics as a Process:

- Statistics is used (and often misused) in a variety of ways.
    - Data can be used to offset anecdotal claims, such as cellular telephones causing brain cancer.
    - We can also show the effectiveness of new drugs, predict changes in house prices, or determine what is associated with determining a house price.
- Why is statistics a field of study?
    - It's all about variability.
    - Look around the class. Does everyone have the same height? Same hair color?
    - Even from person to person, you have a lot of variability. Do you sleep the same number of hours every night? Consume the same number of calories every day?
- One goal of statistics is to describe and understand sources of variability. Statistics itself is a process that has a few steps. We need to define the groups we're analyzing to describe this process.

For many studies, it's not reasonable to access all individuals of interest in your study.



Definition: Population

Definition: Sample

Definition: Individual

Definition: Parameter

Definition: Statistic

Example 1: Suppose the proportion of all students on your campus who have a job is 0.849. Then, suppose a sample of 250 students is obtained, and from this sample we find that the proportion who have a job is 0.864. Identify the population, sample, individuals, parameter, and statistic.

Example 2: In a medical study, researchers wanted to determine if a new drug can lower blood pressure in US adults with high blood pressure. A random sample of 165 US adults with high blood pressure was taken. After 6 months, the drug had lowered blood pressure by an average of 14 points. Identify the population, sample, individuals, parameter, and statistic, if provided in the setup.

The Process of Statistics

1. Identify the research objective.

2. Collect data needed to answer the question(s) above.

3. Describe the data.

4. Perform inference.

Types of Variables

Qualitative (Categorical) vs. Quantitative Variables

Definition: Variable

Key Point: Variables vary. Consider the variable *height*. If all individuals had the same height, then obtaining the height of one individual would be sufficient in knowing the heights of all individuals. Of course, this is not the case. As researchers, we wish to identify the factors that influence variability.

Definition: Qualitative (Categorical) Variables:

Definition: Quantitative Variables:

Example 3: Qualitative versus Quantitative Variables
Classify each of the following variables as qualitative or quantitative.

1. Education level
2. Today's high temperature
3. Daily intake of whole grains (measured in grams per day)
4. Number of vending machines at a school
5. Whether or not a student is prepared for class

Example 4: Qualitative versus Quantitative Variables

Classify each of the following variables as qualitative or quantitative.

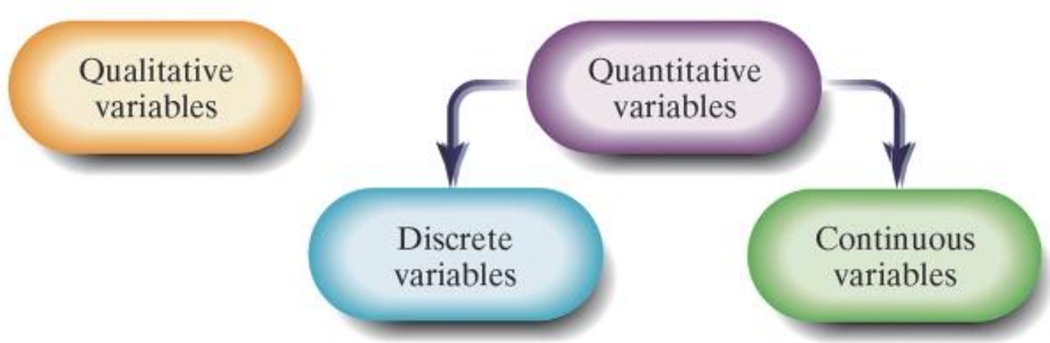
1. Age
2. Generation
3. Student
4. Sleep(hours)

Discrete vs. Continuous Variables

- We can classify quantitative variables further:

Definition: Discrete Variable

Definition: Continuous Variable



Example 5: Discrete Versus Continuous Variables

Classify each of the following variables as qualitative or quantitative. Determine whether the quantitative variables are discrete or continuous.

1. Internet provider

2. Income (in dollars)

3. Grade earned in Algebra (as a percentage)

4. Response to the question "Working with numbers upsets me," (where the responses are given as strongly agree, agree, disagree, or strongly disagree)

5. Number of students in a classroom

Example 6: Discrete Versus Continuous Variables

Classify each of the following variables as qualitative or quantitative. Determine whether the quantitative variables are discrete or continuous.

1. Time Exercised in a day

2. Phone Type (iPhone, Samsung Galaxy, etc.)

3. Temperature, in Celsius

4. Level of Education (Less than High School, High School Diploma, Some College, Bachelor's Degree, Advanced Degree)

5. Number of Newly Admitted Students to UCF

Levels of Measurement

- Another way to classify data.
- Useful for both categorical and quantitative data.

Definition: Nominal Level of Measurement

- If the value of a variable are names, labels, or categories with

Definition: Ordinal level of measurement

- If the value of a variable are names, labels, or categories with

Definition: Interval level of measurement

- If the variable has the properties of the ordinal level but

Definition: Ratio level of measurement

- If the variable has the properties of the interval level but

Example 7: Determining the Level of Measurement of a Variable
For each of the following variables, determine the level of measurement.

1. Internet provider

2. Age (Young, Middle, Old)

3. Age (in years)

4. Response to the question "Working with numbers upsets me," (where the responses are given as strongly agree, agree, disagree, or strongly disagree).

5. Number of students in a classroom

EXAMPLE: Determining the Level of Measurement of a Variable
For each of the following variables, determine the level of measurement.

1. Number of Hours Exercised

2. Phone Type (iPhone, Samsung Galaxy, etc.)

3. Temperature, in Celsius

4. Level of Education (Less than High School, High School Diploma, Some College, Bachelor's Degree, Advanced Degree)

5. Number of Newly Admitted Students to UCF

1.2 Observational Studies Versus Designed Experiments

EXAMPLE: Diet and Energy

Study 1

A researcher wanted to identify the effects of music on intelligence. She identified 68 Kindergarteners and randomly assigned them into three groups for music instruction, focusing on the same songs. Group 1 used xylophone, group 2 used singing, and group 3 listened to music. Lessons continued for 4 months and then IQ tests of the children were administered. A trend was found with the xylophone group and instances of higher IQ.

(Source: Taetle, Laurie Daniels. "The effect of active and passive music instruction on the spatial ability of kindergarten children." (1999))

Study 2

Researchers wanted to identify long-term effects of music on intelligence. An elementary school was identified that held an enrichment program at the after school day-care. The children at the specialized music school outperformed children at a neighboring daycare in cognitive abilities.

(Source: Costa-Giomi, Eugenia. "The long-term effects of childhood music instruction on intelligence and general cognitive abilities." *Update: Applications of Research in Music Education* 33.2 (2015): 20-26)

Both studies have a goal to determine if music has some effect on intelligence.

Definition: Explanatory Variable

Definition: Response Variable

Statistical studies can (generally) be separated into two different groups: Observational Studies and Designed Experiments

Definition: Observational studies

Definition: Designed Experiments

EXAMPLE: Observational Study or Designed Experiment?
Determine whether each of the following studies depict an observational study or an experiment.

1. Researchers studied the prevalence of post-pardum depression (PPD) and associated factors of new mothers. A sample of 279 women aged 19-45 were surveyed in the two months immediately after giving birth. Results showed that unemployment, education status, and C-section were most closely correlated with PPD. (Source: Alzahrani, J., Al-Ghamdi, S., Aldossari, K., Al-Ajmi, M., Al-Ajmi, D., Alanazi, F., ... & Alharbi, A. (2022). Postpartum depression prevalence and associated factors: an observational study in Saudi Arabia. *Medicina*, *58*(11), 1595.)

2. Xylitol has proven effective in preventing dental caries (cavities) when included in food or gum. A total of 75 Peruvian children were given milk with and without xylitol and were asked to evaluate the taste of each. Overall, the children preferred the milk flavored with xylitol. (Source:

Castillo JL, et al (2005) Children’s acceptance of milk with xylitol or sorbitol for dental caries prevention. BMC Oral Health (5)6.)

3. A total of 110 faculty members and 627 students from six engineering departments in the state of California participated in surveys, to identify difficulties experienced during the online instruction. Recommendations for educational reform were made based on past experiences. (Source: Asgari, S., Trajkovic, J., Rahmani, M., Zhang, W., Lo, R. C., & Sciortino, A. (2021). An observational study of engineering online education during the COVID-19 pandemic. *Plos one*, *16*(4), e0250041.)

4. Hypertension (or high blood pressure) can lead to cardiovascular problems and even death. A group of 1006 adolescents with prehypertension (moderately high blood pressure) were tracked over several years to assess which health factors lead to hypertension. (Source: Redwine KM, et al (2011) Development of hypertension in adolescents with pre-hypertension. The Journal of Pediatric, in press.)

Why do we have designed experiments?

- Observational studies do not allow a researcher to claim causation, only association.
    - Observation studies often have **lurking variables,** which are variables not considered in a study, but affect the value of the response variable in the study.
    - Designed experiments may have **confounders,** or variables that were considered in a study whose effect cannot be distinguished from a second explanatory variable in the study.

Other Types of Data Collection

Definition: Census

Definition: Web Scraping (Data Mining)

1.3 Simple Random Sampling

- When we get individuals for a study, we want to select them in a random way so that the results are valid.

Definition: Random Sampling

- The gold standard is a simple random sample.

Definition: Simple Random Sample

Steps for Obtaining a Simple Random Sample

1. Obtain a frame that lists all the individuals in the population of interest. Number the individuals in the frame 1 to N.
2. Use a random number table, graphing calculator, or statistical software to randomly generate n numbers where n is the desired sample size.

EXAMPLE: Obtaining a Simple Random Sample
The House of Representatives has 435 members. Explain how to conduct a simple random sample of 5 members to attend a Presidential luncheon.

EXAMPLE: In 1936, the magazine Literary Digest conducted a poll regarding the outcome of the presidential election between Roosevelt and Landon. They polled 10 million individuals (2.38 million responded) by surveying its own readers, a list of registered automobile owners and that of telephone users.

Explain why they did not use a simple random sample.

1.5 Bias in Sampling

If the results of the sample are not representative of the population, then the sample has **bias.**

There are three main sources of bias:

1. Sampling bias
2. Nonresponse bias
3. Response bias

Definition: Sampling bias

Definition: Nonresponse bias

Definition: Response Bias

Two types of error can be found in samples.

- Nonsampling Errors:

- Sampling Error:

EXAMPLE: According to Martin Boon of ICM Limited, a polling firm in Britain, in 1995 it took 3000 to 4000 calls to obtain a sample size of 2000. Today, it takes over 30,000 calls for the same 2000. To reduce costs, more polling is done using robocalls and internet-based polling. Robocalling to cellular telephones is illegal according to the Telephone Consumer Protection Act (TCPA). How do these methods potentially lead to non-sampling error? What types of non-sampling error do you think this may lead to?

EXAMPLE: In 1936, the magazine Literary Digest conducted a poll regarding the outcome of the presidential election between Roosevelt and Landon. They polled 10 million individuals (2.38 million responded) by surveying its own readers, a list of registered automobile owners and that of telephone users.

They claimed that Landon would win convincingly with about 57.08% of the popular vote. However, Roosevelt won re-election in a landslide and won the popular vote by 24.92 points.

What potential biases led to these poor results?

EXAMPLE: A company wants to determine how well a product is performing. They send out a survey to recent customers and ask them “How satisfied were you with the product?” The answer choices were “Highly satisfied, satisfied, somewhat satisfied, and not satisfied.” What kind of nonsampling errors may occur here?